Last time: Defined standard Δ_λ, costandard ∇_λ } same composition factors (i.e. subquotients of J-H series) assembled in opposite order

Example λ dominant, $w_0 \in W$ longest element ($w_0 \cdot \lambda$ antidominant)

$L_{w_0\cdot\lambda} \hookrightarrow \Delta_\lambda \twoheadrightarrow L_\lambda$ $\qquad$ $L_\lambda \hookrightarrow \nabla_\lambda \twoheadrightarrow L_{w_0\cdot\lambda}$

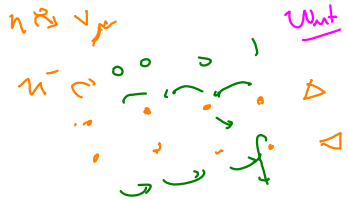
$\Delta_\lambda \to L_\lambda \hookrightarrow \nabla_\lambda$ (unique) $\qquad$ $\nabla_\lambda \to L_{w_0\cdot\lambda} \hookrightarrow \Delta_\lambda$

only projection to simple $\qquad$ only inj. simple $\qquad$ only map $\Delta_\lambda \to \nabla_\lambda$ $\qquad$ $\boxed{\mathrm{Hom}(\nabla_\lambda, \Delta_\lambda) \text{ complicated}}$ $\qquad$ many things to project to $\qquad$ many things to include.

Claim: $\mathrm{Hom}(\Delta_\lambda, \nabla_\mu) = \begin{cases} \mathbb{C} & \text{if } \mu = \lambda \\ 0 & \text{else.} \end{cases}$ } Proof: adjunctions $\mathrm{Hom}(\Delta_\lambda, -) \leftrightarrow$ weight vectors λ killed by $\mathfrak{n}$.

If $\lambda < \mu$: • ∇_μ unique irreducible sub $= L_\mu$.

$n \ni v_\mu$



What weight λ vectors cannot be killed by $\mathfrak{n}$ $\leftarrow$ functional on ∇_λ $\varphi(f, -)$

If $\lambda = \mu$. numerically maximal.

otherwise, $\lambda > \mu$ obvious...

$\lambda < \mu$...

Reason it actually fails numerically highest weight in Δ_λ

λ-space $\cap$ $\ker(\mathfrak{n}) = 0$

Claim: raising operators only kill numerically highest weight

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Rmk: standard filtrations: filtration w/ standard subquot.

costandard ... ... costd - - .

Observation: V finite dim $\qquad - \otimes V$ exact

biadjoint w/ left/right $\qquad = - \otimes V^*$

Prop: Study: $V \otimes \Delta_\lambda$ has a standard filtration. $\qquad V \otimes \nabla_\lambda$

w/ pieces $\Delta_{\lambda + v_i}$ $\qquad$ say V has weights $v_1, \dots, v_r$ (w/ mult.)

"highest" standard is a sub $\qquad \Delta_{\lambda + \max(V)}$ $\qquad$ reversed.

"lowest" standard is quotient $\qquad \Delta_{\lambda + \min(V)}$

Prop: $\mathrm{Hom}_{\mathfrak{g}}(V \otimes \Delta_\lambda, -) \simeq \mathrm{Hom}(\Delta_\lambda, - \otimes V^*)$

$$\simeq \mathrm{Hom}_{\mathfrak{b}}(\mathbb{C}_\lambda, - \otimes V^*)$$

$$\simeq \mathrm{Hom}_{\mathfrak{b}}(\mathbb{C}_\lambda \otimes V, -)$$

$$\simeq \mathrm{Hom}((\mathbb{C}_\lambda \otimes V) \otimes_{U\mathfrak{b}} U\mathfrak{g}, -)$$

Yoneda: $V \otimes \Delta_\lambda \simeq \mathrm{cInd}_{\mathfrak{b}}^{\mathfrak{g}}(\mathbb{C}_\lambda \otimes V)$

filtration: $\mathfrak{b}$-representation.

quotients

sub

Prop. $\mathrm{Ext}^1(\Delta_\lambda, \nabla_\mu) = 0$

$0 \to \nabla_\mu \to M \to \Delta_\lambda \to 0$ split

sub

Proof: · suppose that M has n-faulted λ-vector.

Then splits. ("universal prop. of Δ_λ")

· if not, all λ-weight vectors $= M$ come from ∇_λ

so $\mu > \lambda$.

Apply $(-)^\vee$ $\rightsquigarrow$ $0 \to \nabla_\lambda \to M^\vee \to \Delta_\mu \to 0$.

non-zero $\longleftrightarrow$ v_μ

splits because... $\mu > \lambda$ there is

$\square$

Projectives. **Prop.** $\mathcal{O}$ has enough projectives

· Δ_λ λ dominant projective.

$$\begin{tikzcd} \Delta_\lambda \arrow[d, dashed] \arrow[dr] & \\ M \arrow[r, two heads] & N \arrow[d] \\ & V_\lambda \end{tikzcd}$$

In $\mathcal{O}_\lambda$, every V_λ is n-faulted.

If N has λ-max vector then N does.

$\mathcal{O}_\lambda$

· P projective, V f.d.,

$P \otimes V$ is projective.

$\mathrm{Hom}(P \otimes V, -)$ exact

$\mathrm{Hom}(M, -)$ exact $\Longleftrightarrow$ M projective

· **Show:** surject onto any irreducible.

$$\begin{tikzcd} \Delta_\mu \otimes L_{\mu-\lambda} \arrow[rr, two heads] \arrow[dr, two heads] & & L_\lambda \\ & \Delta_\lambda \arrow[ur, two heads] & \end{tikzcd}$$

earlier $\rightarrow$ smallest shift is quot

μ dominant

$\mu - \lambda$ integral

General theorem:

$$\begin{tikzcd} P_M \arrow[r, two heads] & M \\ P \arrow[u, dashed, "\text{surjective}"] \arrow[ur] & \end{tikzcd}$$

- projective cover: $P_M \twoheadrightarrow M$ projective s.t. every submodule $N \subseteq P_M$ $\mathrm{Im}(N) \neq M$.

- Fact: $\mathcal{C}$ abelian, has projective covers, easier fact: if exist, unique up to iso

Goal:
- finish $\mathcal{O}$
- state main thms of Soergel theory
- start proof, geometric (B-B)